GW170817 confines the vestigial-second-sound graviton identification to one part in 10^{15} of its natural range

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We constrain our assumed leading-order graviton speed $c_{\rm GW} = c\sqrt{\chi_{\rm vest}}$. The conservative GW170817 bound $|\Delta c/c| \le \tau = 3\times10^{-15}$ confines the nonnegative susceptibility to $[(1-\tau)^2, (1+\tau)^2]$, of width 4τ. This interval contains $\chi_{\rm vest} = 1$ and is over fourteen orders of magnitude narrower than our adopted range $[0.1, 10]$. The mode and range are not derived from microscopic dynamics; the constraint applies to this kinematic model, not to emergent gravity generally.

INTRODUCTION

Spinor-gravity constructions use composite vielbeins built from fermions [6, 8]. Volovik proposes a vestigial phase with metric order preceding a nonzero tetrad expectation [1]. His symmetry-breaking analysis distinguishes absorbed Nambu–Goldstone modes from the Higgs-sector gravitational-wave modes, whose counting also uses coordinate constraints [2]. Here we assume that a metric-channel excitation of such a substrate can be identified with a spin-2 second-sound graviton.

Motivation and assumed speed law. Volovik applies Landau two-fluid hydrodynamics to a de Sitter state in [3]. The underlying thermodynamic framework treats the gravitational normal component as stiff matter [4]. Within that de Sitter model, the Landau formula and the stiff-component thermodynamic relations give $s_2 = c$, independently of H. He discusses several possible graviton interpretations of this collective mode and leaves its type open. For the spin-2 substrate identification assumed here, we also assume the susceptibility-dependent speed law:

$$c_{\rm GW}^2(\chi_{\rm vest}) = c^2 \cdot \chi_{\rm vest}, \qquad c_{\rm GW}(\chi_{\rm vest}) = c\sqrt{\chi_{\rm vest}}, \quad (1)$$

where $\chi_{\rm vest}$ is the dimensionless, normalized metric-channel response of the assumed vestigial substrate. It is a free parameter in this calculation. The RPA form in `VestigialSusceptibility.lean` models it using unevaluated bare susceptibility and coupling inputs, so that form alone supplies no numerical value of $\chi_{\rm vest}$. The underlying lattice-fermion framework is the one whose unitarity Vergeles establishes [10]. The identification is appealing in its brevity. In this Letter, Eq. (1) is an assumed kinematic speed law. We constrain its free susceptibility without deriving a propagating second-sound mode from a microscopic action, its propagator, or its coupling to matter. This missing dynamical bridge is a limitation of the present calculation, not an impossibility result.

The 2017 binary-neutron-star merger GW170817 [11] placed an extraordinarily tight bound on $c_{\rm GW}$: comparing the gravitational-wave arrival time at LIGO/Virgo against the gamma-ray-burst signal at Fermi/INTEGRAL, the multi-messenger collaboration constrained $-3 \times 10^{-15} \le \Delta c/c \le +7 \times 10^{-16}$. Conservatively two-sided, $|\Delta c/c| \le \tau$ with $\tau = 3 \times 10^{-15}$.

This Letter formalizes Eq. (1), places it head-to-head against GW170817, and reports the result. The conclusion is sharp: our assumed speed law satisfies the GW170817 bound only on a sub-interval of our adopted range $\chi_{\rm vest} \in [0.1, 10]$ narrower than that range by more than 14 orders of magnitude. The GW170817-compatible window $[(1-\tau)^2, (1+\tau)^2]$ has width 4τ in $\chi_{\rm vest}$ and sits strictly interior to our adopted range $[0.1, 10]$ about the anchor $\chi_{\rm vest} = 1$. Its relative width is $4\tau/9.9 \approx 1.2 \times 10^{-15}$; a probability mass additionally requires a specified prior. The result is a fine-tuning constraint on one specific kinematic identification, not a refutation of emergent gravity nor of the ADW program; we discuss three Phase 6 recovery paths in Sec. .

Relation to the 2017 constraint sweep. Using GW170817 to eliminate a gravity model is not new. Within months of the detection, four back-to-back Letters established what the bound kills: Baker *et al.* [12] covered general scalar-tensor and vector-tensor theories and bounded the graviton mass in bimetric gravity; Creminelli and Vernizzi [13] showed that avoiding tuning requires $c_{\rm GW}$ to be unaffected on *nearby* cosmological solutions, collapsing quartic and quintic beyond-Horndeski theories to a single function; Sakstein and Jain [14] combined the bound with cluster and M87 strong-equivalence-principle data to render the quartic galileon cosmologically irrelevant; and Ezquiaga and Zumalacárregui [15] extended the conclusion to Einstein-aether, Hořava, Generalized Proca and TeVeS.

The present result is not a further entry in that list, and the difference is structural rather than one of degree. Every theory the 2017 sweep reached has a Lagrangian from which a varying $c_{\rm GW}$ is *derived*; the constraint therefore lands on operator coefficients, and each Letter reports which coefficients survive. This Letter supplies no microscopic Lagrangian for the vestigial-second-sound identification. As Sec. notes, $c_{\rm GW} = c\sqrt{\chi_{\rm vest}}$ is assumed kinematically, so this calculation does not constrain microscopic operator coefficients. What the bound constrains is $\chi_{\rm vest}$: the compatible interval has roughly

one part in 10^{15} of the width of our adopted order-unity window. This is an interval-width comparison, not a microscopic prediction of $\chi_{\rm vest}$ or a probability statement independent of a prior. The 2017 sweep constrained *theories*; this constrains an *identification*, which is why it is stated as a ratio of two interval widths rather than as a surviving region of parameter space.

At fixed assumed Eq. (1), this calculation constrains $\chi_{\rm vest}$ rather than microscopic operator coefficients. It does not analyze whether the operator-level mechanisms discussed in those theories alter the effective speed law or its applicability. A microscopic realization would require that separate analysis. The three recovery paths in Sec. can supply a mechanism fixing $\chi_{\rm vest}$ within the compatible interval, or modify the identification or its leading-order assumptions. A fixed-point mechanism can preserve the speed law; the finite-tolerance constraint alone derives none of these mechanisms.

FORMAL SETUP AND RESULT

Vestigial-susceptibility natural range. Our substrate models the vestigial-phase metric-channel susceptibility by the RPA *resummation* $\chi_{\rm RPA} = \chi_0/(1-\gamma_\star\chi_0)$ in the standard Hertz–Millis convention. Here χ_0 is the bubble integral; $\chi_{\rm RPA}$ is the resummed response, not the bubble integral itself. `VestigialSusceptibility.lean` formalizes that closed form and its stability properties; it evaluates neither χ_0 nor $\gamma_\star$, so it fixes no numerical $\chi_{\rm vest}$, which is why $\chi_{\rm vest}$ enters Eq. (1) as a free parameter. The natural range $\chi_{\rm vest} \in [0.1, 10]$ is a *project-adopted* naturalness window: the one-loop bubble normalization is $\mathcal{O}(\Lambda_{\rm UV}^2/16\pi^2)$ by dimensional analysis, and we take the $\pm$1-decade window about the resulting unit-normalized central value. We emphasize the provenance: the window is a modeling choice of this project, not a published derivation: the Vergeles lattice analysis [10] establishes unitarity of the underlying Einstein–Cartan–Palatini lattice theory but does not itself compute $\chi_{\rm vest}$ or assign it a range. The window matches the linearized-EFE wave's ADW microscopic-coefficient $\alpha_{\rm ADW} \in [0.1, 10]$ natural range [16], which we treat as the project-internal anchor for the order-of-magnitude prior.

The leading-order $c_{\rm GW}$. Equation (1) ships in `GravitationalWaves.lean` as `c_GW(chi, c) := c * Real.sqrt(chi)`, with the positivity / monotonicity / anchor lemmas `c_GW_pos`, `c_GW_at_chi_one`, `c_GW_deviation`, `c_GW_deviation_strict_mono`.

The GW170817 correctness-push. Define $\Delta c/c(\chi_{\rm vest}) = \sqrt{\chi_{\rm vest}} - 1$ and the predicate $\texttt{LigoSatisfied}(\delta) \equiv |\delta| \le \tau$. The correctness-push biconditional is

Theorem 1 (`c_GW_match_iff_chi_close_to_one`)**.** *For $\chi_{\rm vest} \ge 0$, $\texttt{LigoSatisfied}(\Delta c/c(\chi_{\rm vest})) \iff \chi_{\rm vest} \in [(1-\tau)^2, (1+\tau)^2]$.*

The proof is a $\sqrt{\cdot}$/squaring round-trip via `Real.sqrt_le_sqrt` and `Real.sqrt_sq`, with `nlinarith` supplied the hint $\sqrt{\chi_{\rm vest}}^2 = \chi_{\rm vest}$. The compatible $\chi_{\rm vest}$ window has width $\sim 4\tau = 1.2 \times 10^{-14}$.

Natural-range fine-tuning. The natural range $[0.1, 10]$ has width 9.9 (`natural_range_width_eq`); the GW170817-compatible window has width exactly $4\tau = 1.2 \times 10^{-14}$ (`ligo_window_width_eq`) and is nested strictly inside it (`ligo_window_subset_natural_range`), not disjoint from it. Scaling the window up by 10^{14} still leaves it the narrower (`ligo_window_width_lt_natural_range_width_by_1e14`): that ratio, not a disjointness, is the fine-tuning claim. Two endpoint-falsifier theorems sharpen the exclusion of the extremes.

Theorem 2 (`natural_lower_violates_ligo`)**.** *At $\chi_{\rm vest} = 0.1$, $\Delta c/c = \sqrt{0.1} - 1 \approx -0.6838$, hence $|\Delta c/c|/\tau \approx 2.28 \times 10^{14}$ and* `LigoSatisfied` *is false.*

Theorem 3 (`natural_upper_violates_ligo`)**.** *At $\chi_{\rm vest} = 10$, $\Delta c/c = \sqrt{10} - 1 \approx +2.1623$, hence $|\Delta c/c|/\tau \approx 7.21 \times 10^{14}$ and* `LigoSatisfied` *is false.*

The proofs use $\sqrt{1/10} \le \sqrt{1/4} = 1/2$ and $\sqrt{10} \ge \sqrt{4} = 2$, both via `Real.sqrt_le_sqrt` and `Real.sqrt_sq`, closed by `linarith`. The Lean source supplies proofs of these two concrete real-number inequalities. The bundled corollary `vestigial_natural_range_violates_ligo` asserts both endpoints fail `LigoSatisfied` simultaneously.

The identification hypothesis itself is refuted pointwise rather than only at the observable: `H_VestigialModeIsGraviton_fails_at_natural_lower` and `H_VestigialModeIsGraviton_fails_at_natural_upper` refute it at the two endpoints, and `H_VestigialModeIsGraviton_fails_at_zero` and `H_VestigialModeIsGraviton_fails_at_quarter` refute it at $\chi_{\rm vest} \in \{0, 1/4\}$. Zero lies outside the adopted range, whereas 1/4 is an interior point. These falsifiers show failures away from the two natural-range endpoints, without excluding the compatible interval. `ligo_satisfied_at_chi_one` supplies the compatible anchor $\chi_{\rm vest} = 1$. The exact-zero statement `c_GW_deviation_zero_iff_chi_one` characterizes vanishing deviation, not finite-tolerance compatibility: the latter holds on the entire interval in Theorem 1.

Tracked-hypothesis bundle. Per the project's discipline for non-derived identifications, we bundle the structural properties of "the vestigial mode is the graviton"

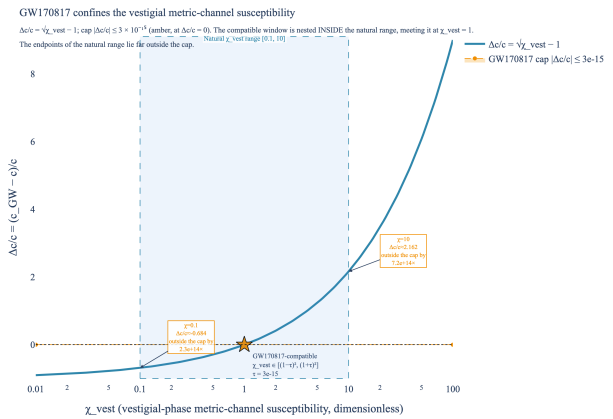


FIG. 1. **Headline result.** The $\Delta c/c = \sqrt{\chi_{\rm vest}} - 1$ deviation across the natural $\chi_{\rm vest} \in [0.1, 10]$ range (pale blue band, dashed blue border) versus the GW170817 [11] multi-messenger cap $|\Delta c/c| \leq 3 \times 10^{-15}$ (amber, drawn at $\Delta c/c = 0$ because the cap's own width is far below one pixel at this scale). Over the natural range $\Delta c/c$ sweeps $[-0.68, +2.16]$, exceeding the cap by $\sim 10^{14}\times$ *at the endpoints* (orange callouts) and vanishing at $\chi_{\rm vest} = 1$. The compatible $\chi_{\rm vest}$ window $[(1-\tau)^2, (1+\tau)^2]$ has width $4\tau = 1.2 \times 10^{-14}$, compared with the adopted range's width 9.9. Its finite extent is unresolved at this scale; the gold star marks its anchor $\chi_{\rm vest} = 1$, not a singleton compatible set. The window sits *inside* the band: the datum confines $\chi_{\rm vest}$, it does not exclude the range. Lean witnesses: `ligo_confines_chi_vest_within_natural_range`, `ligo_window_subset_natural_range`, `vestigial_natural_range_violates_ligo`.

into a Lean Prop with three recorded conjuncts:

$$\begin{aligned} &\texttt{H_VestigialModeIsGraviton}(\chi_{\rm vest}) \\ &\quad \equiv \chi_{\rm vest} > 0 \\ &\qquad \wedge \texttt{LigoSatisfied}(\Delta c/c(\chi_{\rm vest})) \\ &\qquad \wedge |\Delta c/c(\chi_{\rm vest})| < 1/2, \end{aligned}$$

demanding positivity (P1), GW170817 compatibility (P2), and sub-half-deviation $|\Delta c/c| < 1/2$ (P3'). These clauses are not independent at the fixed tolerance $\tau = 3 \times 10^{-15}$: P2 already implies P3' because $\tau < 1/2$, and the compatible interval in Theorem 1 is strictly positive, supplying P1 throughout that interval. The recorded Prop retains all three clauses; this observation does not change its definition.

The bundle is satisfied at $\chi_{\rm vest} = 1$ (`H_VestigialModeIsGraviton_at_one`). Four explicit falsifier theorems establish that it is not universally satisfied: `*_fails_at_natural_lower` and `*_fails_at_natural_upper` use failure of P2; `*_fails_at_zero` uses failure of P1; `*_fails_at_quarter` uses failure of P3'. At $\chi_{\rm vest} = 1/4$, $\Delta c/c = -1/2$ fails both P2 and P3', while satisfying P1. It therefore shows that P1 alone does not imply P3', not that P3' is independent of the other two clauses. Likewise these proofs do not isolate one failing clause while requiring all others to hold. On the nonnegative physical domain, the bundle's compatible set is precisely the finite interval in Theorem 1, of width $4\tau = 1.2 \times 10^{-14}$. Its probability mass depends on the prior; under a uniform prior on $[0.1, 10]$ it is $4\tau/9.9$.

Endpoint constraints and the open dynamical bridge. The theorem `natural_endpoints_violate_ligo_window` excludes $\chi_{\rm vest} \in \{0.1, 10\}$ under the assumed speed law. The relative-deviation form `natural_endpoints_violate_ligo_for_every_c` uses `c_GW_relative_deviation_eq` and holds for every $c \neq 0$, including the physical domain $c > 0$. Both concern the two endpoints only; neither excludes interior values, and $\chi_{\rm vest} = 1$ satisfies the cap (`ligo_satisfied_at_chi_one`). These are conditional constraints on the assumed law. Neither theorem quantifies over microscopic actions, propagators or matter couplings, so neither establishes the absence or impossibility of a dynamical derivation.

DISCUSSION

What this constrains. The conservative symmetric cap restricts the assumed leading-order identification to $\chi_{\rm vest} \in [(1-\tau)^2, (1+\tau)^2]$. Expanding the endpoints gives $1 - 2\tau + \tau^2$ and $1 + 2\tau + \tau^2$: the susceptibility offsets are approximately $\pm 6 \times 10^{-15}$, whereas the speed-deviation tolerance is $\pm 3 \times 10^{-15}$. Exact luminality occurs at $\chi_{\rm vest} = 1$, but the finite measurement tolerance permits a nonzero-width interval. Its small relative width within our adopted range does not alone establish a prior probability, rule out generic UV theories, or require an exact fixed point at unity.

What this does not rule out. Three Phase 6 paths remain open:

1. **Symmetry-driven $\chi_{\rm vest} \to 1$ fixed point.** If a deep-IR symmetry (e.g. Lorentz invariance of the emergent metric sector) makes $\chi_{\rm vest} = 1$ an exact fixed point of the RG flow, the fine-tuning is structural rather than tuned. An exact fixed point at $\chi_{\rm vest} = 1$ would lie inside the compatible interval. The finite-tolerance constraint alone does not select this mechanism or establish that a microscopic model realizes it. Phase 6 should examine whether such a fixed point is realized in the ADW broken phase.

2. **Distinct metric-channel identification.** We propose as an alternative that the "vestigial mode" is a separate UV-input metric channel rather than the second-sound DOF. Its dynamics could in principle pin its susceptibility to unity; this remains a model-building possibility. The missing dynamical bridge in this Letter does not establish that this

alternative is more likely. Phase 6e (full nonlinear emergent action) is the natural venue for the disambiguation.

3. **Frequency-dependent $\chi_{\rm vest}$.** Equation (1) is an assumed frequency-independent kinematic law. A frequency-dependent susceptibility or a compensating correction in the LIGO band is a possible phenomenological extension. We consider the algebraic ansatz $\delta_{\rm disp} = \gamma\omega$, without asserting that it is the leading term of a matched microscopic derivative expansion. The formal probe is numerical $\omega = 100$; interpreting it as a 100 Hz band and setting $\gamma = \Gamma_H/c_{\rm GW}^2$ are modeling assumptions requiring physical normalization and microscopic matching. `vestigial_dispersion_below_ligo_at_inspiral_peak` assumes $|\gamma| \le 10^{-30}$ and proves that $|\delta_{\rm disp}|$ at that numerical probe is below the GW170817 tolerance. The adopted 10^{-30} is a placeholder chosen to keep the correction below the cap; it is neither derived nor measured. In the further additive model $\delta_{\rm tot} = \delta_0 + \delta_{\rm disp}$, with $\delta_0 = \sqrt{\chi_{\rm vest}} - 1$, that assumed coefficient bound gives $|\delta_{\rm disp}| \le B = 10^{-28}$ at $\omega = 100$. The reverse triangle inequality then excludes restoration of $|\delta_{\rm tot}| \le \tau$ whenever $|\delta_0| > \tau + B$. Offsets just outside $\pm\tau$ can nevertheless become compatible through an oppositely signed correction within this bound. Larger corrections are not excluded by the placeholder theorem. For positive ω, `dispersion_within_ligo_iff` exchanges the correction's tolerance for $|\gamma| \le {\rm tol}/\omega$. `dispersion_correction_abs_bound`, `dispersion_correction_linear_in_gamma` and `dispersion_correction_zero_at_no_dissipation` describe the product's magnitude, additivity in γ and zero-coefficient limit. Deriving Γ_H and the normalization of this ansatz from microscopic dynamics remains open.

Comparison to the linearized-EFE wave. The companion linearized-EFE wave [16] ships an emergent-gravity identification *compatible* with observation at the natural-range level: the Sakharov–Adler form $G_{\rm N,emerg} = \alpha_{\rm ADW} \cdot 12\pi/(N_f \Lambda_{\rm UV}^2)$ at $\Lambda_{\rm UV} = M_{\rm Pl}$ produces $\alpha^\star_{\rm ADW} \in [0.40, 1.27]$ for SM N_f, sitting inside the natural range $[0.1, 10]$. The asymmetry between the two waves is a feature of the program: the linearized-EFE $G_{\rm N,emerg}$ depends on the Sakharov–Adler kernel computed at the linearized level, where the relevant scale is $M_{\rm Pl}$; $c_{\rm GW}$ depends on the free metric-channel susceptibility $\chi_{\rm vest}$. Our adopted interval contains unity but derives no value of this susceptibility. The conservative GW170817 cap restricts it to the much narrower compatible interval; the width comparison alone supplies no microscopic selection mechanism.

Reproducibility. The result is reproducible in two layers. The Lean layer is `GravitationalWaves.lean` (32 theorems, zero `sorry`, zero new axioms, zero `maxHeartbeats` overrides) at the project's pinned Mathlib4 commit; the Python layer is `src/gravitational_waves/` with 81 `pytest` cases cross-checking every numeric quoted in the abstract and figure caption. The headline figure (`src/core/visualizations.py:fig_c_GW_vs_ligo_constraint`) and its underlying numerics are committed-pinned.

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